Name :

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Vellore Institute of Technology
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Continuous Assessment Test I – January 2025

Programme	:	B.Tech. (CSE)	Semester	:	Winter 2024-25
Course	:	Embedded Systems	Code	:	BCSE305L
			Class Nbr	:	CH2024250501701 CH2024250501716 CH2024250501718 CH2024250501712
Faculty	:	Prof. Satheesh Kumar T, Dr. Sritama Roy, Dr. Manigandan M, Dr. Sridhar C	Slot	:	G2+TG2
Time	:	90 Minutes	Max. Marks	:	50

Answer ALL the questions

Q.No.	Sub. Sec.	Questions	Marks	BT Level
1.		Describe the design process of smart watering system, including sensor placement, data processing, and communication system. Draw a neat block diagram to explain the hardware and software architecture and also explain the different phases involved in the above design.	10	2
2.		With a neat Sketch of the architecture describe the microcontroller runs on reduced instruction set computer optimized for managing peripherals and control-oriented tasks in embedded systems.	10	2
3.		Apply an appropriate embedded system design process to design a face detection based Biometric Attendance System. List out its essential features and write appropriate code for the design.	10	3
4.		In a petrochemical industry having boiler for generating steam, you need to develop a system to monitor the temperature of the boiler. The following conditions are crucial for monitoring the boiler. i) Temperature > 600°C – Display an alert message "Boiler is Overheating!" in serial monitor and glow red LED. Also turn On the cooling system. ii) Temperature < 450°C (idle state) – Display "Idle State" in serial monitor and glow yellow LED. iii) Temperature between 450°C and 600°C – Display " Boiler Working" in serial monitor and glow green LED Write an Arduino program to develop an optimal embedded system for the petrochemical plant.	10	3

5.	Write an Arduino program to design a smart parking system using HC-SR04 ultrasonic sensor, servo motor, buzzer and Arduino Uno. ➢ The ultrasonic sensor module placed near the entry gate continuously checks for the incoming vehicles. ➢ When a vehicle comes closer to the ultrasonic sensor detection area and parking slot is available then the system opens a gate barrier to 90° (close after 10 seconds) to allow the vehicle to the parking slot and decrements the parking slot by 1. ➢ If no parking slot is available, switch on the buzzer for 5 seconds. Have a similar system on the exit and increment the free slot by 1 for every vehicle which leaves the parking slot. Note: Total capacity of the parking slot is 10.	**10**	**3**